



Public-Key Encryption



Public-key / Asymmetric Encryption

Asymmetric encryption (or asymmetric key encryption):

- **One key** is used for encryption (usually publicly known, *public key*);
- **Another key** is used for decryption (usually *private*, or *secret key*)

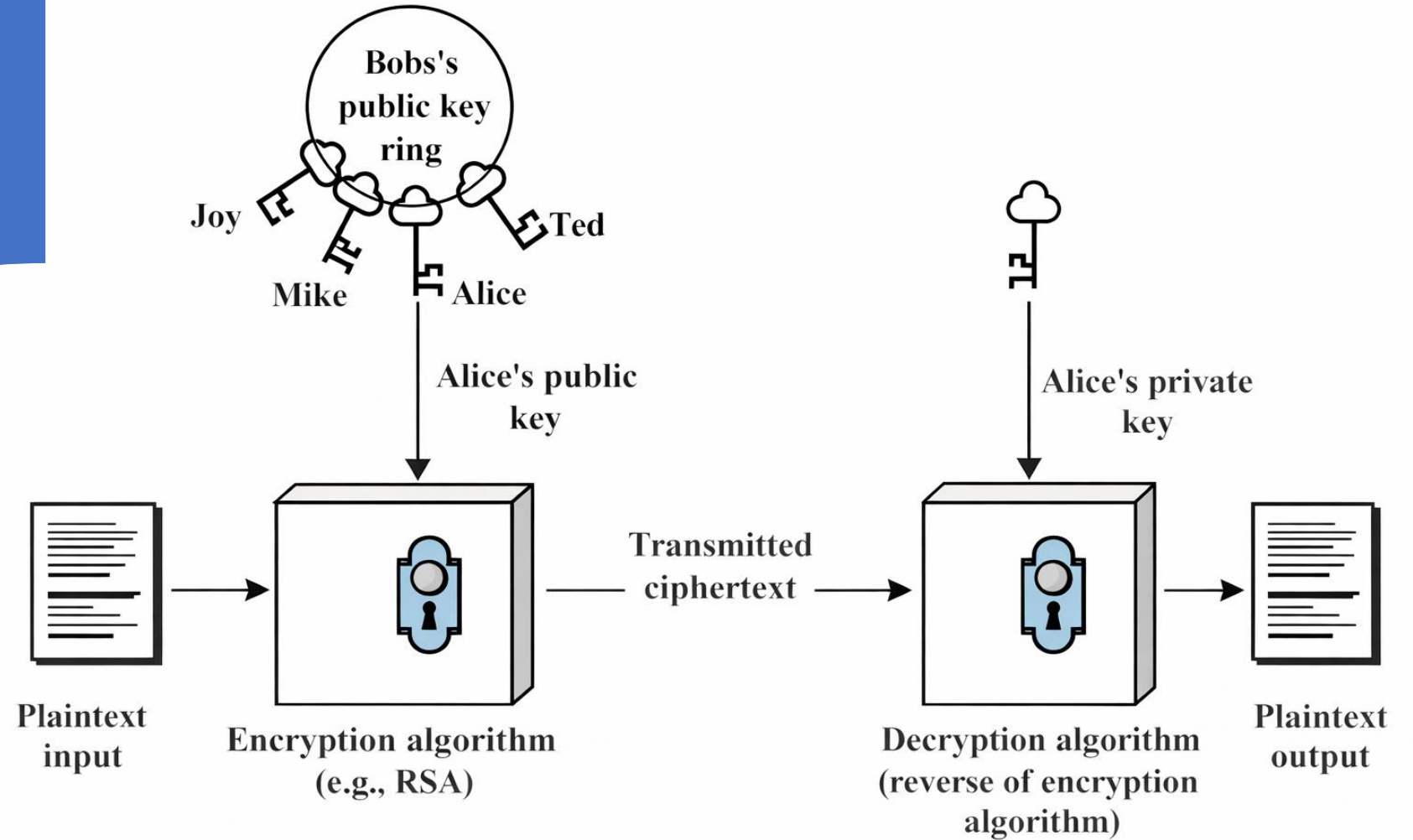
Alice and Bob agree on a public key algorithm.

Alice sends Bob the public key. (or key is retrieved from a repository)

Bob encrypts a message with Alice's public key and sends it to Alice.

Alice decrypts Bob's message with the private key.

Public-key encryption



(a) Encryption

Components of public-key encryption

Plaintext

Encryption algorithm (RSA / ECDSA / DSA / DH)

Public and private key

Ciphertext

Decryption algorithm

A close-up photograph of a hand holding a set of keys. The keys are silver and attached to a metal ring. The background is blurred, showing what appears to be a person's face and some indoor lighting. The image is partially obscured by a white curved shape on the right side, which contains the text.

Public-key encryption

Advantages

- All keys (public and private) are generated locally
- No need in secure distribution of the keys
- Moreover, each user can change his own pair of public/private key at any time

Disadvantages

- It is more computationally expensive

Applications of Public-Key Cryptosystems



Encryption/decryption:

the sender encrypts a message with the recipient's public key.



Digital signature (authentication):

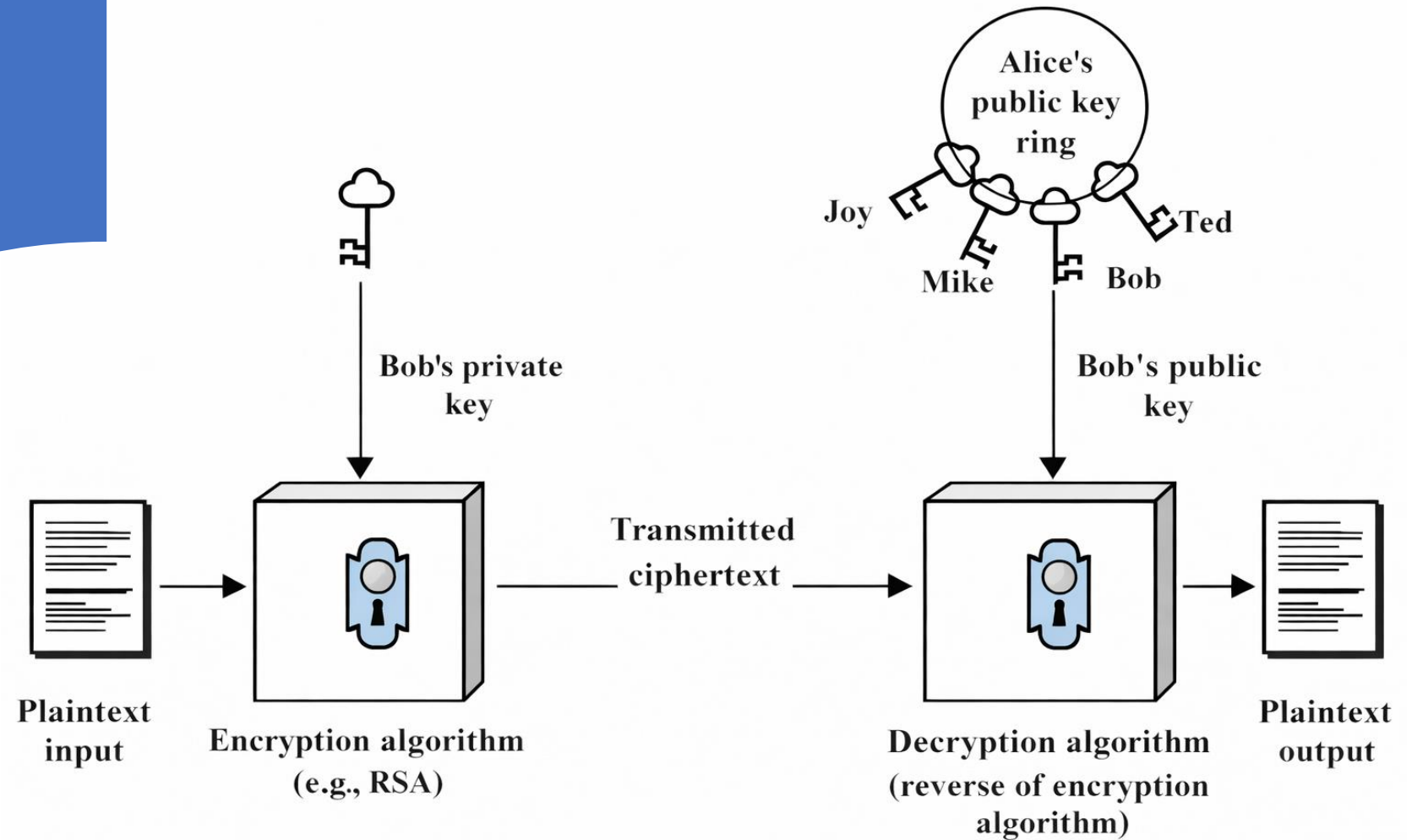
the sender "signs" the message with its private key; a receiver can verify the identity of the sender using sender's public key.



Key exchange:

both sender and receiver cooperate to exchange a (session) key.

Authentication using public-key systems



(b) Authentication

RSA allows for verification



Only the private key holder can generate a message that can be decrypted by the public key



A secure method of authenticating the owner of the private key is the person who sent the message

A close-up photograph of a hand holding a set of keys. The hand is positioned on the left side of the frame, with the thumb and index finger gripping a keyring. The keys are silver and have a modern, rectangular design. The background is blurred, showing a person's face and a dark sleeve, suggesting an indoor setting.

Two-way secure communication

- A signs with A's **private key** and then *encrypts* with B's **public key**
 - This gives us authenticity because only A's private key could have produced the signature
 - B's private key is the only key that can unlock the other encryption
- No one other than the owners of these two private keys could view this message
- Any modification would invalidate the signature
- Sign for authenticity, integrity. Encrypt for confidentiality.